

FIXME Introduces a reminder that this part of the code doesn't entirely work

HACK Introduces a reminder that this part of the code works, perhaps barely, but that the code should be improved

xxx Introduces a general warning, often of high severity

You should follow these always-uppercase labels with more specific descriptions of the task or problem at hand. Later, you can search the source code for the labels to find the code that needs fixing. The downside is that you can easily forget about these reminders unless you happen to be reading the section of your code that they're in. Codetags shouldn't replace a formal issue tracker or bug reporter tool. If you do use a codetag in your code, I recommend keeping it simple: use only `TODO` and forgo the others.

Magic Comments and Source File Encoding

You might have seen `.py` source files with something like the following lines at the top:

```
❶ #!/usr/bin/env python3
❷ # -*- coding: utf-8 -*-
```

These *magic comments*, which always appear at the top of the file, provide interpreter or encoding information. The shebang line ❶ (introduced in Chapter 2) tells your operating system which interpreter to use to run the instructions in the file.

The second magic comment is an *encoding definition* line ❷. In this case, the line defines UTF-8 as the Unicode encoding scheme to use for the source file. You almost never need to include this line, because most editors and IDEs already save source code files in the UTF-8 encoding, and Python versions starting with Python 3.0 treat UTF-8 as the defined encoding by default. Files encoded in UTF-8 can contain any character, so your `.py` source file will be just fine whether it includes English, Chinese, or Arabic letters. For an introduction to Unicode and string encodings, I highly recommend Ned Batchelder's blog post, "Pragmatic Unicode" at <https://nedbatchelder.com/text/unipain.html>.